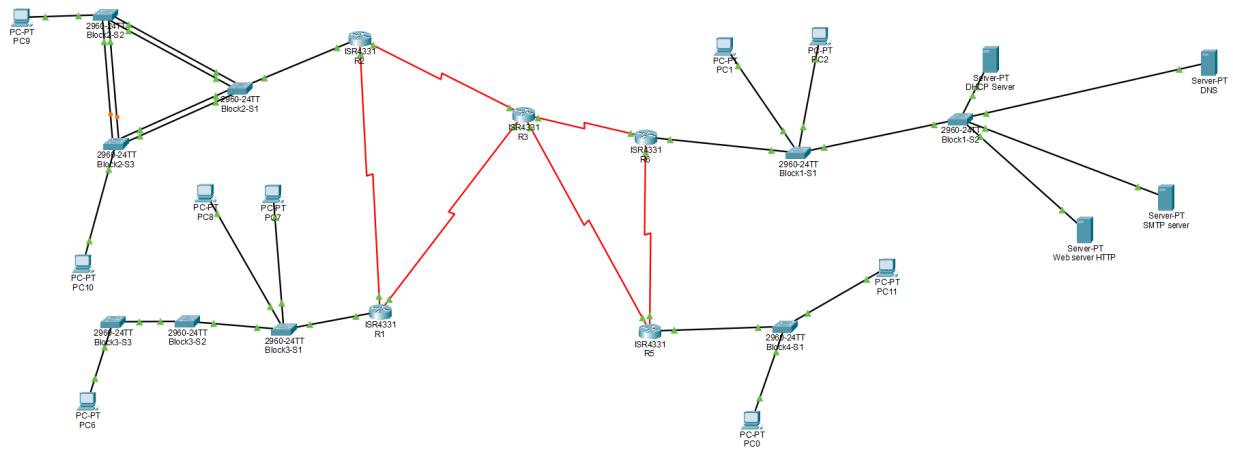
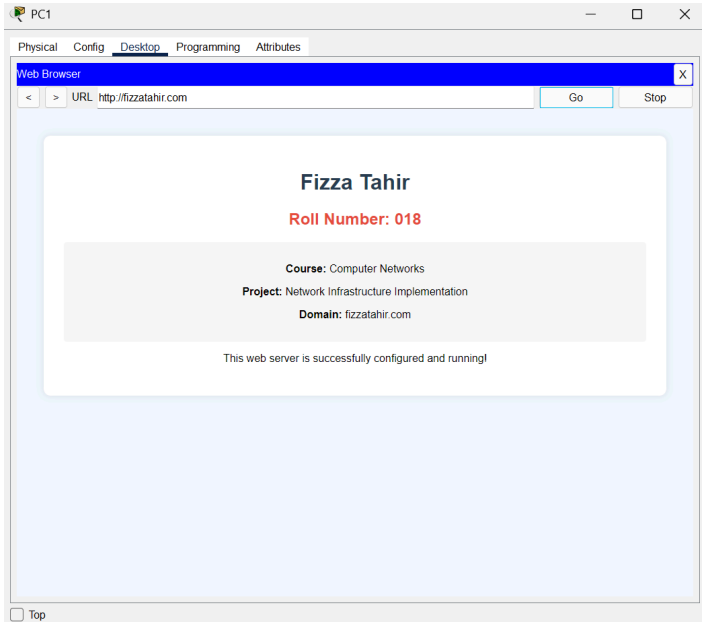
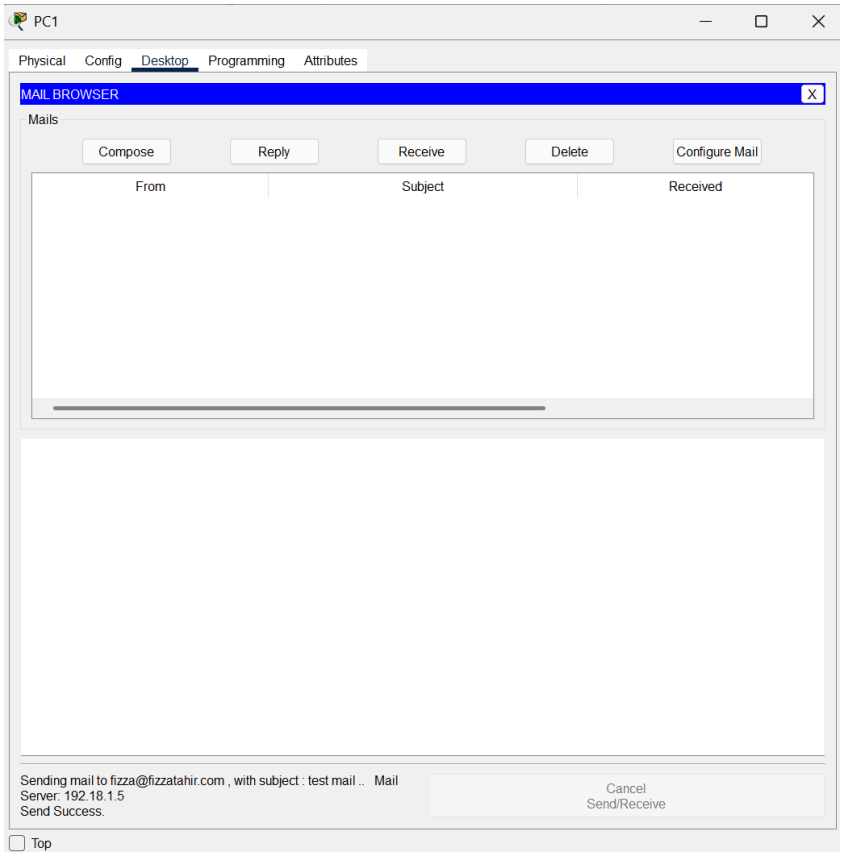
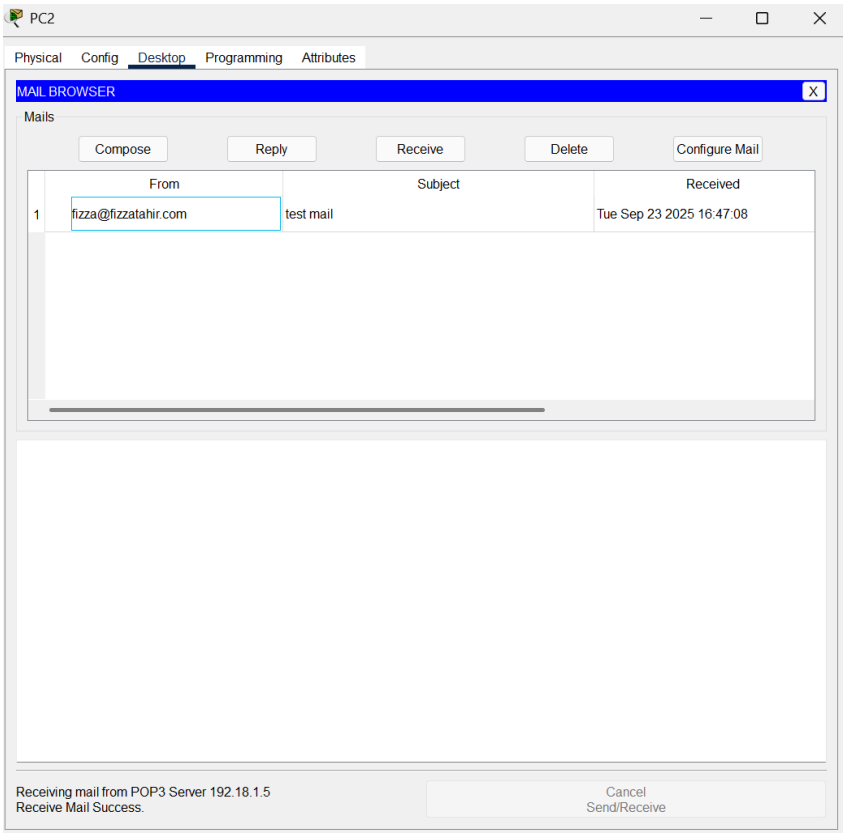
Computer Networks
Project-1
BSAI24018
FIZZA TAHIR
Topology:



Task 1:




PC1
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
Sending mail to fizza@fizzatahir.com , with subject : test mail .. Mail Server: 192.18.1.5
Send Success.
Cancel
Send/Receive
Top


PC2
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
1
fizza@fizzatahir.com
test mail
Tue Sep 23 2025 16:47:08
Receiving mail from POP3 Server 192.18.1.5
Receive Mail Success.
Cancel
Send/Receive

Task 2:
PC 9 to R2 and PC10

```
C:\>ping 192.18.10.1

Pinging 192.18.10.1 with 32 bytes of data:

Reply from 192.18.10.1: bytes=32 time<1ms TTL=255
Reply from 192.18.10.1: bytes=32 time<1ms TTL=255
Reply from 192.18.10.1: bytes=32 time<1ms TTL=255
Reply from 192.18.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.18.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.18.10.11

Pinging 192.18.10.11 with 32 bytes of data:

Reply from 192.18.10.11: bytes=32 time<1ms TTL=128
Reply from 192.18.10.11: bytes=32 time<1ms TTL=128
Reply from 192.18.10.11: bytes=32 time<1ms TTL=128
Reply from 192.18.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.18.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC10 to R2 and PC9

```
C:\>ping 192.18.10.1

Pinging 192.18.10.1 with 32 bytes of data:

Reply from 192.18.10.1: bytes=32 time<1ms TTL=255
Reply from 192.18.10.1: bytes=32 time<1ms TTL=255
Reply from 192.18.10.1: bytes=32 time<1ms TTL=255
Reply from 192.18.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.18.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.18.10.10

Pinging 192.18.10.10 with 32 bytes of data:

Reply from 192.18.10.10: bytes=32 time<1ms TTL=128
Reply from 192.18.10.10: bytes=32 time=1ms TTL=128
Reply from 192.18.10.10: bytes=32 time<1ms TTL=128
Reply from 192.18.10.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.18.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Task 3:

```
C:\>telnet 192.18.40.10
Trying 192.18.40.10 ...Open


User Access Verification

Password:
Block3-S1>
Block3-S1>exit

[Connection to 192.18.40.10 closed by foreign host]
C:\>telnet 192.18.40.11
Trying 192.18.40.11 ...Open


User Access Verification

Password:
Block3-S2>exit

[Connection to 192.18.40.11 closed by foreign host]
C:\>telnet 192.18.40.12
Trying 192.18.40.12 ...Open


User Access Verification

Password:
Block3-S3>
```

Task 4:

PC11

```
C:\>ping 192.18.4.1

Pinging 192.18.4.1 with 32 bytes of data:

Reply from 192.18.4.1: Destination host unreachable.
Reply from 192.18.4.1: Destination host unreachable.
Reply from 192.18.4.1: Destination host unreachable.
Reply from 192.18.4.1: Destination host unreachable.

Ping statistics for 192.18.4.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC0

```
C:\>ping 192.18.4.1

Pinging 192.18.4.1 with 32 bytes of data:

Reply from 192.18.4.1: bytes=32 time<1ms TTL=255
Reply from 192.18.4.1: bytes=32 time=1ms TTL=255
Reply from 192.18.4.1: bytes=32 time=16ms TTL=255
Reply from 192.18.4.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.18.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 16ms, Average = 4ms
```

Task 5:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.18.10.10

Pinging 192.18.10.10 with 32 bytes of data:

Request timed out.
Reply from 192.18.10.10: bytes=32 time=2ms TTL=125
Reply from 192.18.10.10: bytes=32 time=3ms TTL=125
Reply from 192.18.10.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.18.10.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>ping 192.18.20.10

Pinging 192.18.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.18.20.10: bytes=32 time=13ms TTL=125
Reply from 192.18.20.10: bytes=32 time=2ms TTL=125
Reply from 192.18.20.10: bytes=32 time=3ms TTL=125

Ping statistics for 192.18.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 13ms, Average = 6ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.18.11.10

Pinging 192.18.11.10 with 32 bytes of data:

Reply from 192.18.11.10: bytes=32 time=2ms TTL=125
Reply from 192.18.11.10: bytes=32 time=2ms TTL=125
Reply from 192.18.11.10: bytes=32 time=2ms TTL=125
Reply from 192.18.11.10: bytes=32 time=10ms TTL=125

Ping statistics for 192.18.11.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 10ms, Average = 4ms

C:\>
```

Task 6:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.18.1.3     # Web Server in Block1
Invalid Command.

C:\>ping 192.18.1.3

Pinging 192.18.1.3 with 32 bytes of data:

Request timed out.
Reply from 192.18.1.3: bytes=32 time=1ms TTL=126
Reply from 192.18.1.3: bytes=32 time=3ms TTL=126
Reply from 192.18.1.3: bytes=32 time=2ms TTL=126

Ping statistics for 192.18.1.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>
```

Web server HTTP

Physical | Config | Services | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.18.4.10

Pinging 192.18.4.10 with 32 bytes of data:

Reply from 192.18.4.10: bytes=32 time=1ms TTL=126
Reply from 192.18.4.10: bytes=32 time=1ms TTL=126
Reply from 192.18.4.10: bytes=32 time=2ms TTL=126
Reply from 192.18.4.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.18.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```

Task 7:
Pinging from pc9 to webserver and pc0

```
C:\>ping 192.18.1.3

Pinging 192.18.1.3 with 32 bytes of data:

Reply from 192.18.1.3: bytes=32 time=3ms TTL=124
Reply from 192.18.1.3: bytes=32 time=25ms TTL=124
Reply from 192.18.1.3: bytes=32 time=6ms TTL=124
Reply from 192.18.1.3: bytes=32 time=4ms TTL=124

Ping statistics for 192.18.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 25ms, Average = 9ms

C:\>ping 192.18.4.10

Pinging 192.18.4.10 with 32 bytes of data:

Reply from 192.18.4.10: bytes=32 time=5ms TTL=124
Reply from 192.18.4.10: bytes=32 time=30ms TTL=124
Reply from 192.18.4.10: bytes=32 time=3ms TTL=124
Reply from 192.18.4.10: bytes=32 time=3ms TTL=124

Ping statistics for 192.18.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 30ms, Average = 10ms
```

Pinging from pc0 to pc9 and pc6

```
C:\>ping 192.18.10.10

Pinging 192.18.10.10 with 32 bytes of data:

Reply from 192.18.10.10: bytes=32 time=20ms TTL=125
Reply from 192.18.10.10: bytes=32 time=11ms TTL=125
Reply from 192.18.10.10: bytes=32 time=2ms TTL=125
Reply from 192.18.10.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.18.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 20ms, Average = 8ms

C:\>ping 192.18.11.10

Pinging 192.18.11.10 with 32 bytes of data:

Reply from 192.18.11.10: bytes=32 time=3ms TTL=124
Reply from 192.18.11.10: bytes=32 time=3ms TTL=124
Reply from 192.18.11.10: bytes=32 time=3ms TTL=124
Reply from 192.18.11.10: bytes=32 time=3ms TTL=124

Ping statistics for 192.18.11.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3ms, Average = 3ms
```

Task 8:
Pc0 to any router

```
R1>telnet 192.18.4.1
Trying 192.18.4.1 ...Open


User Access Verification

Password:
R5>enable
Password:
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    192.18.1.0/24 [110/65] via 192.18.50.2, 00:41:41, Serial0/1/0
     192.18.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.18.4.0/24 is directly connected, GigabitEthernet0/0/0
L       192.18.4.1/32 is directly connected, GigabitEthernet0/0/0
O E2 192.18.10.0/24 [110/20] via 192.18.51.2, 00:59:19, Serial0/1/1
O E2 192.18.11.0/24 [110/20] via 192.18.51.2, 00:14:50, Serial0/1/1
O E2 192.18.20.0/24 [110/20] via 192.18.51.2, 00:59:19, Serial0/1/1
O E2 192.18.30.0/24 [110/20] via 192.18.51.2, 00:59:19, Serial0/1/1
O E2 192.18.40.0/24 [110/20] via 192.18.51.2, 00:59:19, Serial0/1/1
     192.18.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.18.50.0/30 is directly connected, Serial0/1/0
L       192.18.50.1/32 is directly connected, Serial0/1/0

R5#exit

[Connection to 192.18.4.1 closed by foreign host]
```

Pc6 to r1 direct and r1 vlan

```
C:\>telnet 192.18.10.1
Trying 192.18.10.1 ...Open


User Access Verification

Password:
R1>telnet 192.18.20.1
Trying 192.18.20.1 ...Open


User Access Verification

Password:
R1>
```

Pc9 to r2 direct connection

```
C:\>telnet 192.18.11.1
Trying 192.18.11.1 ...Open


User Access Verification

Password:
R2>
```